

Analysis Report : Database Design for Last Resort Hotels

Group 2

Team Members:

Bulat Iglamov, Jasmine Zhu, Yucheng Ming, Elshadie Solomon

Assumptions:

1. Adjacency is recorded only when two rooms are physically connected by a door, this includes standard doors, private-access doors, and movable walls with doors.
2. Adjacency is symmetric, so the same room pair should not be stored twice in reverse order.
3. One room may be adjacent to multiple other rooms.
4. An event is a scheduled gathering that may use one or more rooms.
5. Each event has one responsible host/customer.
6. Each event has one billing party responsible for payment.
7. The table stores event-level details only, not room-specific assignment details.
8. A single event may use multiple rooms.
9. A single room may be used by multiple events over time.
10. Each row represents one event-to-room assignment.
11. The event_room_assignment table stores room-specific usage details for an event, such as usage type, duration, or charge.
12. event_room_assignment table resolves the many-to-many relationship between event and room.
13. Smoking Status is conditional on the wing, not the floor.
14. A single stay may involve multiple billing parties.
15. Billing charges could be split among different customers.
16. Messages may originate from internal customers or external callers.

Last Resort Hotels:

The easiest approach to create ERD and Tables for this assignment is to go from general – hotel complexes and buildings – to specific – rooms and features. Hence, our intuition starts us out with creating a unique identifying table for hotel complex.

hotel_complex		
hotelComplexId (PK)	name	address
1	Hotel Star	100 Park Avenue, New York, NY
2	Hotel Dream	200 Park Avenue, New York, NY
3	Hotel Delight	300 Park Avenue, New York, NY

Recognizing that each hotel complex has multiple buildings, and each building has multiple wings, we need to create a separate entity for building and wing as follows.

building		
buildingID (PK)	name	hotelComplexId (FK)
1	Earth	1
2	Venuse	1
3	Mercury	2

wing					
wingId (PK)	name	buildingId (FK)	proximityFeatures	handicappedAccess	smokingPolicy
1	Wind	1	Close to indoor swimming pool	True	Smoking is prohibited
2	Rain	1	Close to parking garage	True	Smoking is prohibited
3	Sun	1	Close to outdoor swimming pool and parking garage	False	Smoking is permitted only in specialized areas

The next step is to understand the categorization of the room entity. The main concern is that multiple rooms can have multiple features, and multiple features can be in multiple rooms, leading to many-to-many relationships that we need to resolve. We will resolve this by creating a separate entity room_feature_assignment (as shown below). With the current set up – that room entity will only comprise necessary characteristics – it would be easier for us to resolve room adjacency in the future, as well as dealing with events. Below you can find three entities: room,

room_feature_assignment and room_feature. Important Note: entity relationships will be presented at the conclusion of the analysis.

room									
roomId (PK)	wingId (FK)	number	floor	type	maxOccupancy	baseRate	smokingPolicy	hasToilet	status
1	1	11	1	Sleeping	4	100	Not Permitted	True	Available
2	2	21	2	Sleeping	4	120	Not Permitted	True	Under Renovation
3	3	31	3	Suite	6	250	Not Permitted	True	Available

room_feature_assignment		
roomId (PK, FK)	featureId (PK, FK)	quantity
1	1	2
2	3	1
3	3	2

feature		
featureId (PK)	name	value
1	Bed Type	King Size
2	Bed Type	Queen Size
3	Rollaway Beds	1 Rollaway Beds

Our next goal is to understand 1) how to structure room adjacency and 2) understand the dynamic between customer, event and room and why we need to use a bridge table to assign an event with room and customer with room (explained later).

The next step is to determine whether rooms are adjacent to one another. This is best modeled as a many-to-many relationship, because one room may be connected to multiple other rooms, while each adjacency links a pair of rooms. Accordingly, room_adjacency serves as a bridge table on room, with (roomId, roomId) as a composite primary key to prevent duplicate pairings and represent each valid connection clearly. Examples of adjacency in the case include standard door connections, private-access doors between sleeping rooms, and movable walls with doors.

room_adjacency		
roomId(PK, FK)	roomId(PK, FK)	adjacencyType
1	2	moveable wall
3	4	door
5	6	separate
1	7	private access door
2	5	moveable wall

In this business structure, customer represents the person or organization responsible for hosting or being associated with an event, event represents the gathering, and event_room_assignment is the table that connects events to the physical rooms they use. This means the relationship is that one customer can be associated with many events, and each event can use one or many rooms, while each room can be used by many different events over time. Therefore, event_room_assignment captures the many-to-many relationship between event and room, along with event-specific details such as usage, duration, and charges(related to billing_arrangement).

event						
eventId(PK)	roomId(FK)	customerId(FK)	billingId(FK)	duration	estimatedAttendance	eventName
100	1	121	100	2 hours	100	art forum
101	2	122	101	4 hours	60	math forum
102	3	123	102	3 hours	90	lunch

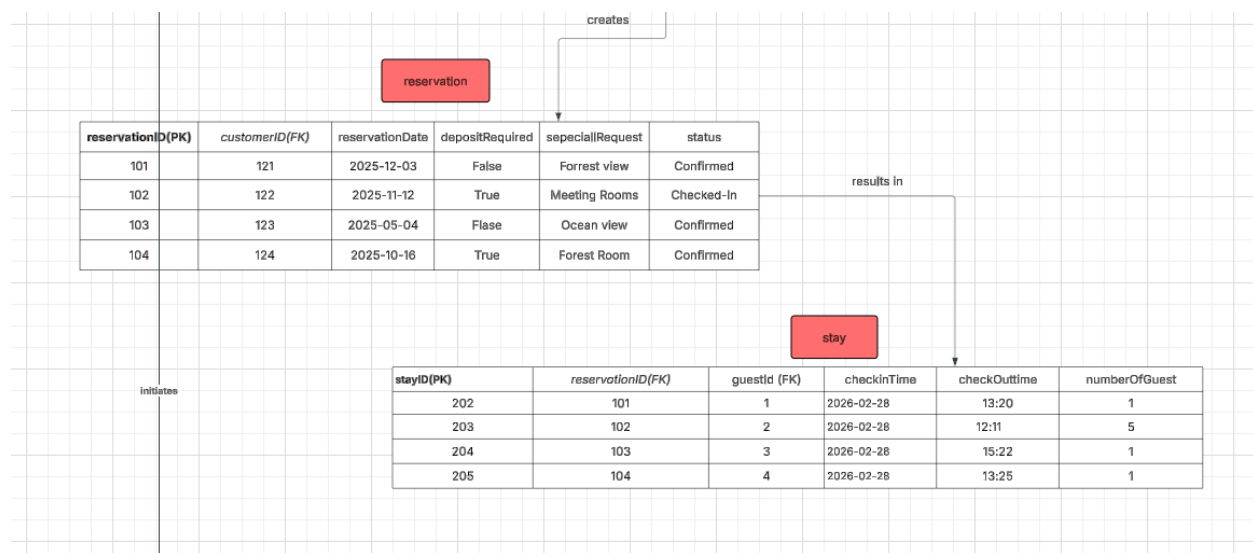
event_room_assignment				
eventId(PK, FK)	roomId(PK, FK)	duration	usage	charge
100	1	2 hours	meeting	100.00
101	2	4 hours	meeting	120.00
102	3	3 hours	lunch	300.00

As we approach from the side of the customer, the first move is to create a customer information table. We are able to recognize the customer table where it serves as a directory of every individual and organization that has a connection with Last Resort Hotels. It acts as a table that stores the core information about all the customers, including the customerID, Type and contact details. The customer type is able to distinguish between individuals or organizations (companies etc). As well as the unique customerID serving as the primary key to identifying each customer throughout the reservation. This is the foundation of the system as it would be able to connect to every reservation and billing arrangement.

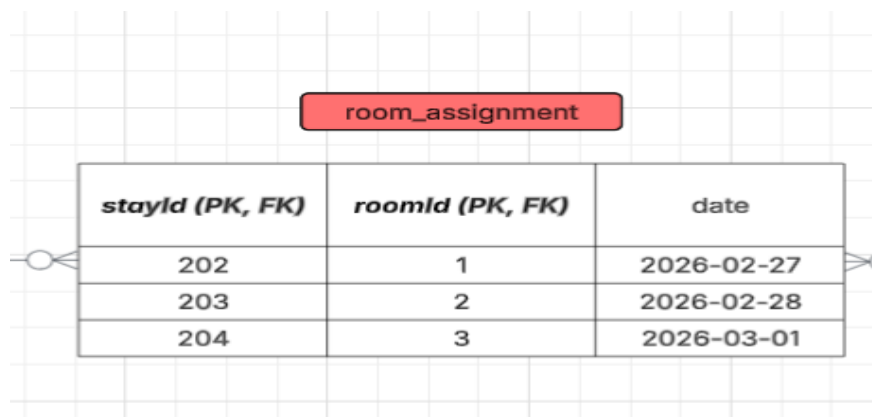
customer					
customerID(PK)	customerType(FK)	firstName	lastName	organizationName	email
121	Individual	Katy	Smith	NULL	Katy..Smith@email.com
122	organization	John	Wilson	Eventcorp	events.corp@email.com
123	individual	Mary	White	NULL	Mary.W@email.com
124	individual	Joseph	Harris	NULL	J.Harris@email.com

Furthermore, the reservation table serves as a booking record that highlights when a customer reserves rooms within the hotel. The unique reservationID is able to identify each booking throughout the system. The customerID is able to show which specific customer went ahead and booked which room. While the reservation date tracks when the specific booking was made. A special request serves to show the customer preferences and which the status of their reservation was.

The next step would be to set up the stay table as it represents the physical occupation of a room by the guest. The StayID serves to show each specific room and how it links to reservations through the (reservationID). This shows which physical room each guest is in and includes the checkInTime, and checkOutTime. The number of guests shows how many people are staying in that room and serves as a bridge between reservation and the actual room assignment.



In addition, the room assignment is able to state which specific room is given to the specific guest at that time. The unique roomID shows each individual room assignment in the hotel . As well as the stayID to show which specific guest stays in this assignment. Including the reserved date for that specific room.



Then connecting it to the employee table which indicates the staff members that work at the Last Resort hotel. Their employeeID shows which department they are in (deptID) , which also includes their first and last name. As well as their typical work schedule that tracks which part of the day they are in for.

employee					
employeeID(PK)	deptID(FK)	firstName	lastName	position	shift
326	10	James	Brown	Front Desk Manager	day
327	11	Lisa	Jay	Event coordinator	day
328	12	Michael	Garcia	Housekeeper	day

message						
messageId (PK)	senderName	senderCustomerId (FK)	messageContext	timeRecorded	status	confidential
400	Katy Smith	121	Late Check Out	2026-02-28 13:25	Read	False
401	John Wilson	122	Event Arrangement	2026-02-28 14:56	Read	False
402	Front Desk	Null	Package Delivery	2026-02-28 15:25	Delivered	False
403	Unknown Caller	Null	Context Main Office Urgently	2026-02-28 16:00	New	True

Message is an important communication tool. Since staff or guests might receive calls or notifications from external parties, the system must store these messages in an organized and secure manner, and also explains why senderCustomerId is added along with senderName. Confidentiality is significant in the table as it ensures privacy and safety.

access_log						
2026-02-28	customerId (FK)	facilities (FK)	accessDate	accessTime	direction	accessMethod
300	121	Lounge	2026-02-28	15:00	Enter	Card
301	123	Gym	2026-02-28	15:30	Enter	Card
300	121	Lounge	2026-02-28	16:15	Exit	Card
302	122	Meeting Room	2026-02-28	16:30	Enter	Pin
301	123	Gym	2026-02-28	16:55	Exit	Card
302	122	Meeting Room	2026-02-28	19:30	Exit	Pin

Other than billing information, access is also an important operational component of the hotel system since guests and staff use either access cards or PIN codes to enter rooms and facilities.

Recording the time they entered and exited is also important as a way to record actions rather than storing the different current locations.

Billing is an important component of the hotel system because guests and event organizers may use multiple rooms, facilities, and services during their stay. The billing structure must therefore track charges from different services and associate them with the appropriate customer or billing party.

billing_arrangement				
billingId (PK)	stayId (FK)	billedPartyId (PK,FK)	billingType	splitPercent
100	202	121	Room	100.00
101	203	122	Incidentals	50.00
102	204	123	Room	50.00
103	205	124	Incidentals	100.00

Billing is modeled at the stay level so the system can distinguish which customer is responsible for paying for different charges based on the services and facilities they used. StayId shows at what specific time and phrase the billing occurs, and billed party Id counts on the customer in response to the billing.

service			
serviceId(PK)	serviceName	serviceType	defaultRate
1	Room Over Night	Internal	400.00
2	Bar Charge	Internal	0.00
3	Restaurant Charge	Internal	0.00
4	Airport Shuttle	External	40.00
5	Meeting Room Rental(per hour)	Internal	100.00

The service table lists the services provided by the hotel and associates each service with a unique serviceId. Default rate represents the standard unit price for services that have fixed pricing. However, some services such as bars and restaurants may not have a fixed rate because their charges depend on the specific items or orders made by customers.

transaction						
transactionId (PK)	billingId(FK)	serviceId(FK)	date	time	finalAmount	authorizationStatus
201	100	1	2026-02-28	15:00	200.00	Posted
202	100	2	2026-02-28	20:00	24.50	Posted
203	101	1	2026-02-28	15:00	200.00	Posted
204	102	1	2026-02-28	15:00	200.00	Posted
205	103	1	2026-02-28	15:00	200.00	Posted
206	102	3	2026-02-28	19:30	78.50	Authorized
207	102	4	2026-02-28	07:30	40.00	Posted
208	103	5	2026-02-28	10:00	100.00	Authorized

The transaction table records each charge incurred during a stay or event by linking a billingId with a serviceId. The timestamp records when the charge occurred. By associating transactions with billing arrangements, the system can allocate charges to the correct billing parties and ensure that the final bill is calculated accurately at checkout.

Tradeoffs and limitations:

1. event_room_assignment may be too simplified if room usage needs detailed time slots, overlaps, or different charges by period.
2. room_adjacency only shows that rooms are connected, but may not fully describe the details, type or temporary nature of the connection. It also doesn't account for any changes in the type of adjacency over time.
3. It does not fully capture more complex real-world rules such as changing event room status, flexible room use, or advanced scheduling constraints.
4. event_room_assignment may be inaccurate in practice, because room uses such as breakfast or meeting can change, while the table only captures the scheduled or recorded assignment.
5. billing_arrangement assumes splitPercent can be calculated through simple percentage allocations. When interpreting real-world data, there would be much more complex structure which this design cannot capture completely.
6. transaction records only record charges at service level. In reality there would be breakdown details such as individual items and partial payments.
7. access_log records only enter and exit for the cards and pins. This does not record how many people and who exactly gets into the facilities as cards and pins may share among different customers and guests.

Handling ambiguities:

1. The ambiguity of whether adjacency should be stored once or twice was resolved by assuming adjacency is symmetric and storing each room pair only once.
2. The article describes an event as a gathering that may use one or more rooms, but does not clearly define whether room information should be stored directly in the event table. This was handled by treating it as an event-level table only. The event table was assumed to store general event details, specific room usage is handled in event_room_assignment.
3. The case explains that an event may use multiple rooms and that rooms may be reused over time, but does not clearly specify how detailed scheduling and usage records should be. This was handled by creating event_room_assignment as a junction table between event and room. Each row was assumed to represent one event-to-room assignment. Attributes such as usage, duration, and charge were included as practical assumptions to capture room-specific event details.

Work Division:

Bulat Iglamov: hotel information and room-feature-assignment, connecting tables, formatting the analysis report, handling assumptions.

Jasmine Zhu: billing information and related tables, handling messages, ambiguities and tradeoffs and limitations.

Yucheng Ming: room adjacency and assignment, connecting tables, handling tradeoffs and limitations.

Elshadie Solomon: customer information and related tables, formatting, handling assumptions for customer information.